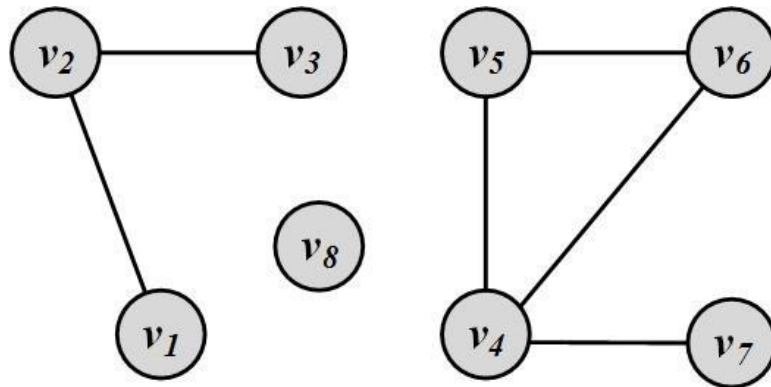


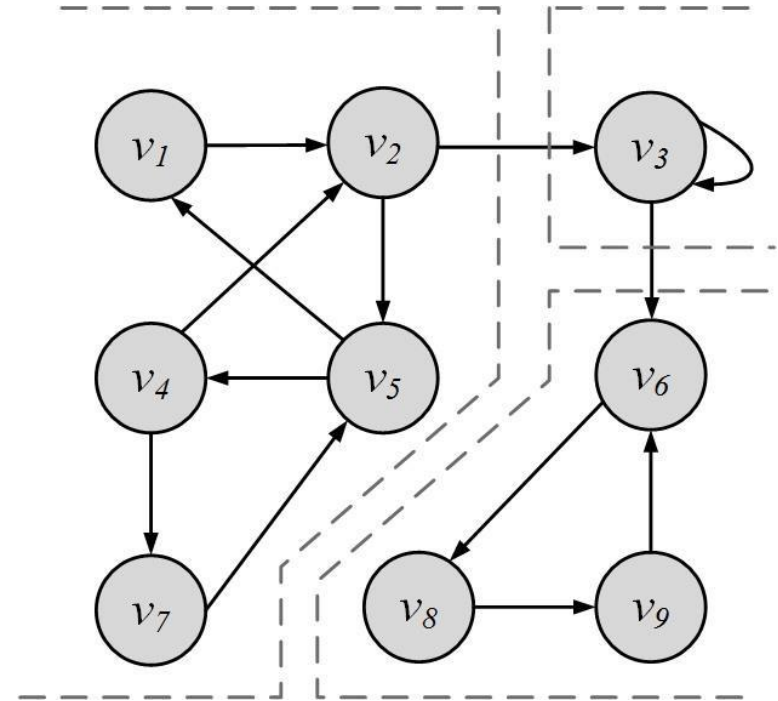
Component

- A **component** in an undirected graph is a connected subgraph, i.e., there is a path between every pair of nodes inside the component
- In directed graphs, we have a **strongly connected** components when there is a path from u to v and one from v to u for every pair of nodes u and v .
- The component is **weakly connected** if replacing directed edges with undirected edges results in a connected component

Component Examples:



3 components



3 Strongly-connected components

Shortest Path

- **Shortest Path** is the path between two nodes that has the shortest length.
 - We denote the length of the shortest path between nodes v_i and v_j as $l_{i,j}$
- The concept of the neighborhood of a node can be generalized using shortest paths. An **n-hop neighborhood** of a node is the set of nodes that are within n hops distance from the node.

Diameter

The diameter of a graph is the length of the longest shortest path between any pairs of nodes in the graph

$$\textit{diameter}_G = \max_{(v_i, v_j) \in V \times V} l_{i,j}$$

- How big is the diameter of the web?

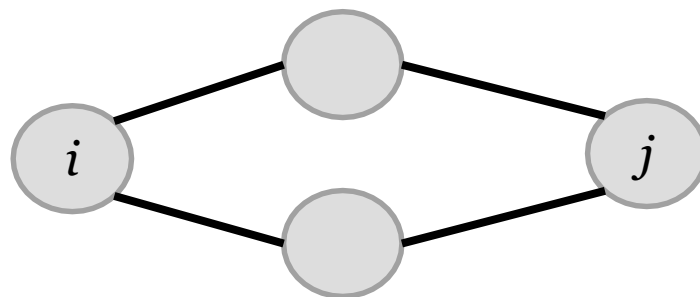
Adjacency Matrix and Connectivity

- Consider the following adjacency matrix

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} & \dots & A_{1n} \\ A_{21} & A_{22} & A_{23} & \dots & A_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ A_{d1} & A_{d2} & A_{d3} & \dots & A_{dn} \end{bmatrix}$$

- Number of Common neighbors between node i and node j

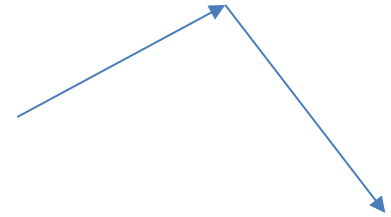
$$\sum_k A_{ik} A_{jk} = A_i \cdot A_j$$



- That's element of $[ij]$ of matrix $A \times A^T = A^2$
- The number of common neighbors with path of length 2

Adjacency Matrix and Connectivity

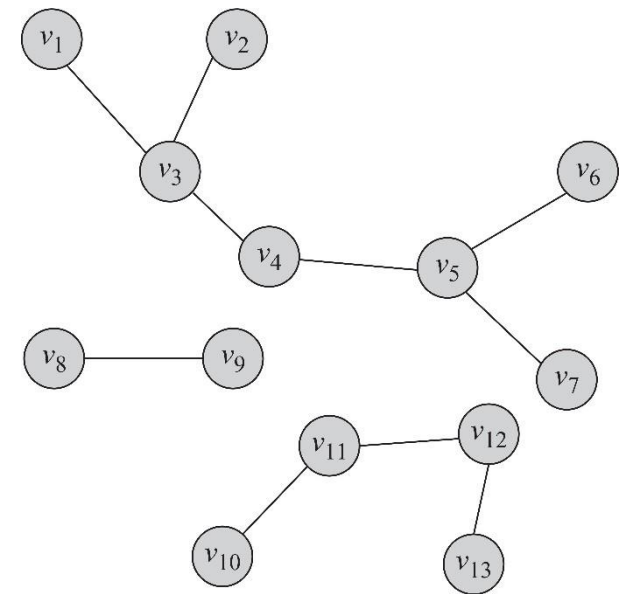
- For either directed or undirected graph
- $A_{ij} = 1$, if there is an edge from i to j
- $A_{ik}A_{kj} = 1$, if there is a path of length 2 from i to j via k
- The total number of paths of length 2 from i to j , via any node, $N_{ij}^{(2)} = \sum_{k=1}^n A_{ik}A_{kj} = [A^2]_{ij}$
- Similarly, $N_{ij}^{(3)} = \sum_{k,l} A_{ik}A_{kl}A_{lj} = [A^3]_{ij}$, the total number of paths of length 3



Special Graphs

Trees and Forests

- **Trees** are special cases of undirected graphs
- A tree is a graph structure that has no cycle in it
- In a tree, there is exactly one path between any pair of nodes
- In a tree: $|V| = |E| + 1$
- A set of disconnected trees is called a **forest**



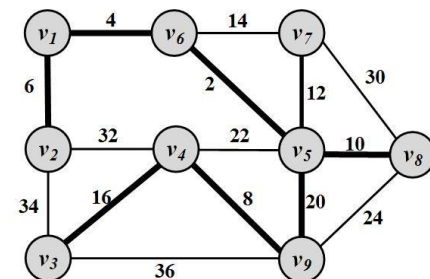
A forest containing 3 trees

Special Subgraphs

Spanning Trees

- For any connected graph, the spanning tree is a subgraph and a tree that includes all the nodes of the graph
- There may exist multiple spanning trees for a graph.
- In a weighted graph, the weight of a spanning tree is the summation of the edge weights in the tree.
- Among the many spanning trees found for a weighted graph, the one with the minimum weight is called the

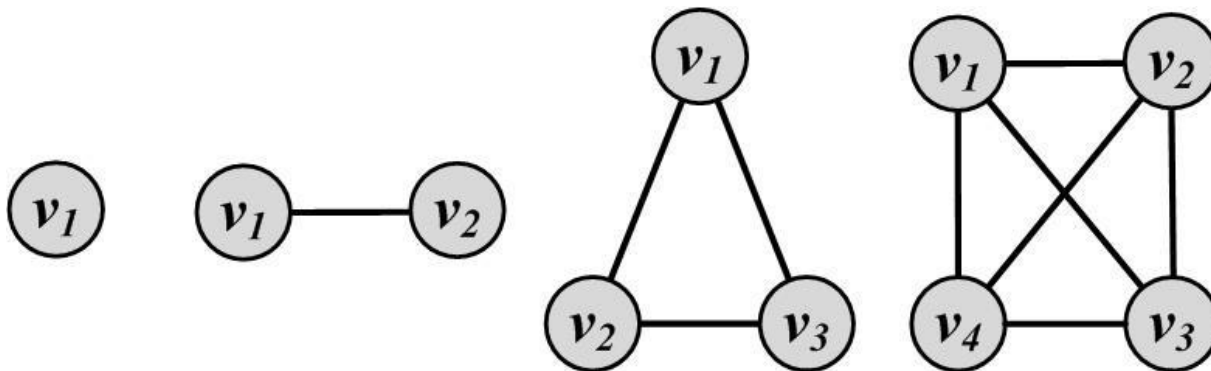
minimum spanning tree (MST)



Complete Graphs

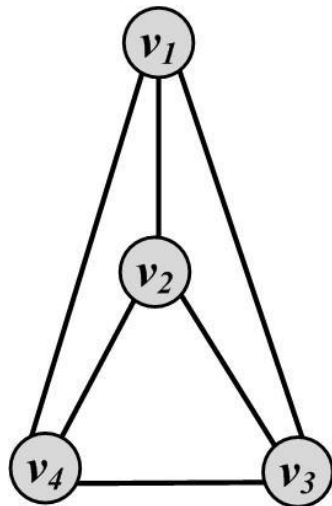
- A complete graph is a graph where for a set of nodes V , all possible edges exist in the graph
- In a complete graph, any pair of nodes are connected via an edge

$$|E| = \binom{|V|}{2}$$

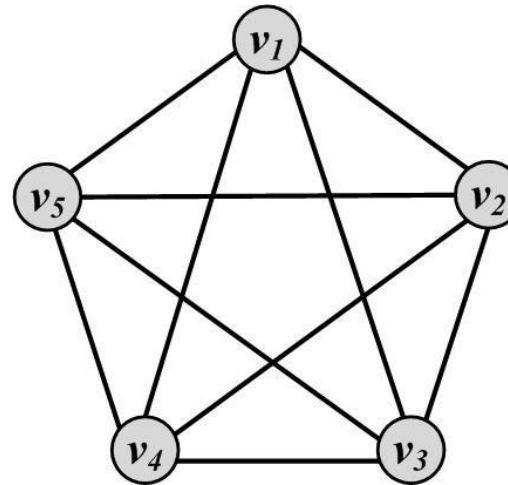


Planar Graphs

A graph that can be drawn in such a way that no two edges cross each other (other than the endpoints) is called planar



Planar Graph

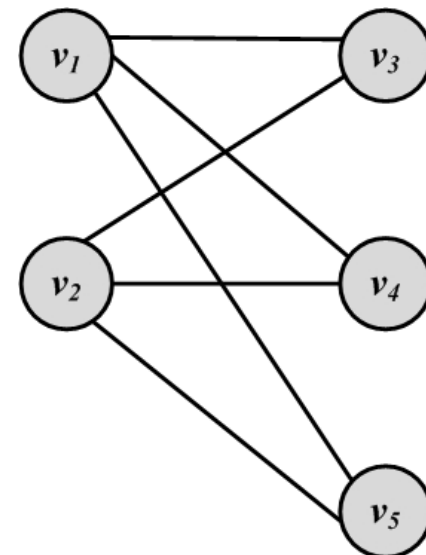


Non-planar Graph

Bipartite Graphs

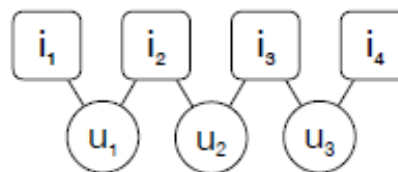
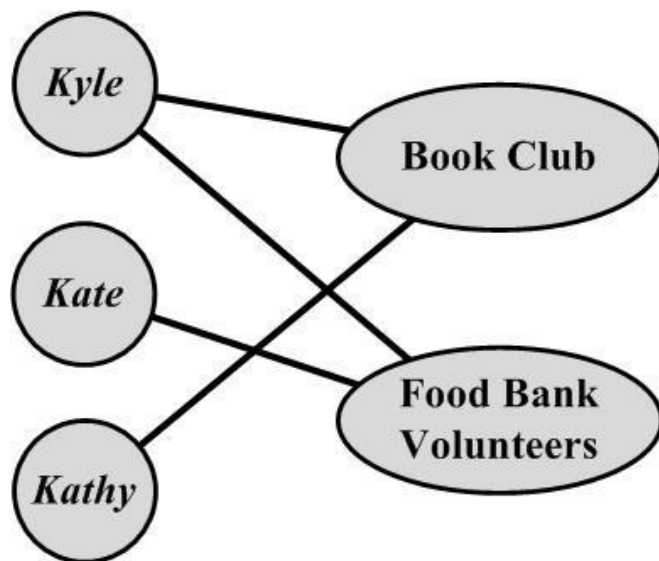
A bipartite graph $G(V, E)$ is a graph where the node set can be partitioned into two sets such that, for all edges, one end-point is in one set and the other end-point is in the other set.

$$\begin{cases} V = V_L \cup V_R, \\ V_L \cap V_R = \emptyset, \\ E \subset V_L \times V_R. \end{cases}$$



Affiliation Networks

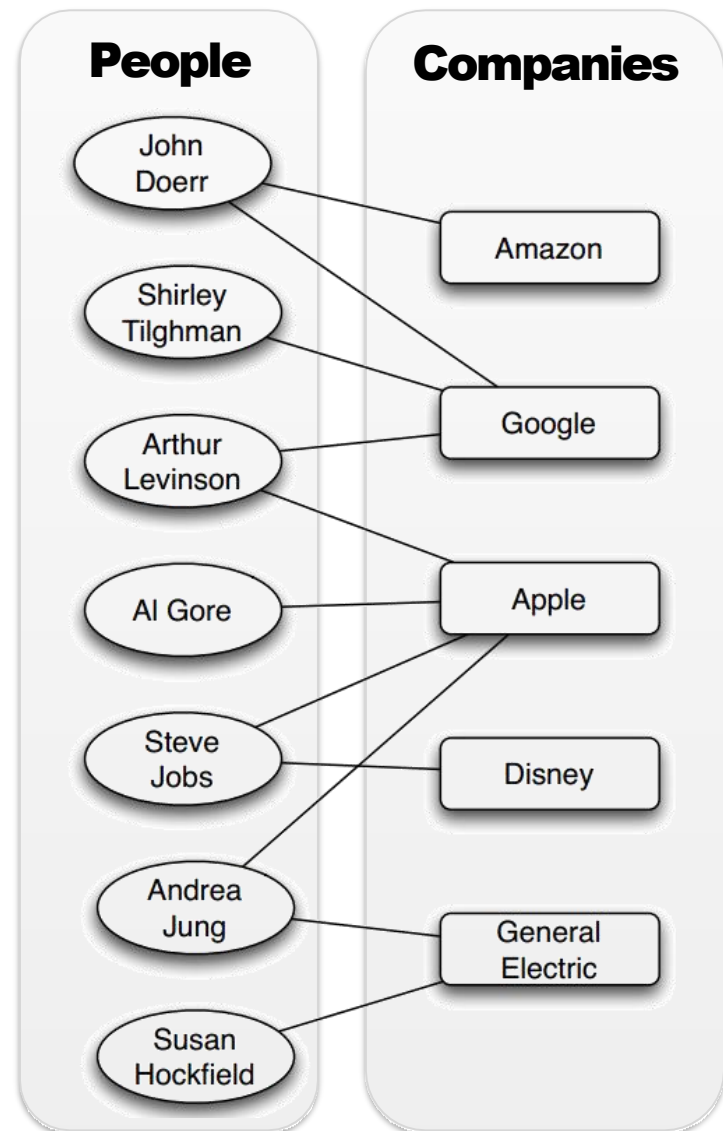
An affiliation network is a bipartite graph. If an individual is associated with an affiliation, an edge connects the corresponding nodes.



User-item interaction
like, share, purchase, ratings etc.

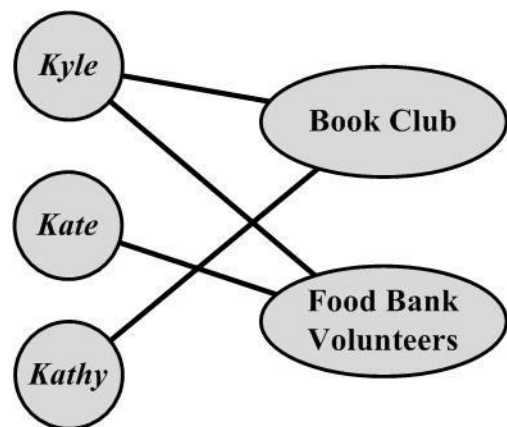
Affiliation Networks: Membership

Affiliation of people on corporate boards of directors



Bipartite Representation / one-mode Projections

- We can save some space by keeping membership matrix X



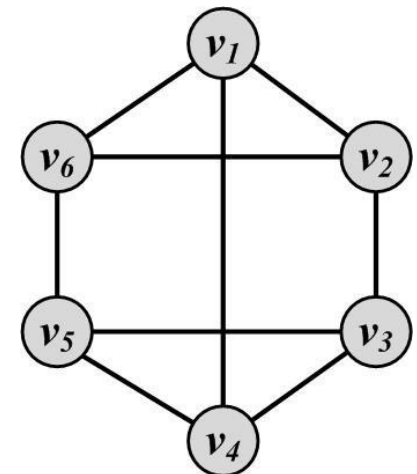
$$X = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- What is XX^T ? *Similarity between users*
- What is X^TX ? *Similarity between groups (items)*

Elements on the diagonal are number of groups
the user is a member of
OR
number of users in the group

Regular Graphs

- A regular graph is one in which all nodes have the same degree
- Regular graphs can be connected or disconnected
- In a k -regular graph, all nodes have degree k
- Complete graphs are examples of regular graphs



Regular graph
With $k = 3$

Graph/Network Traversal Algorithms

Graph/Tree Traversal

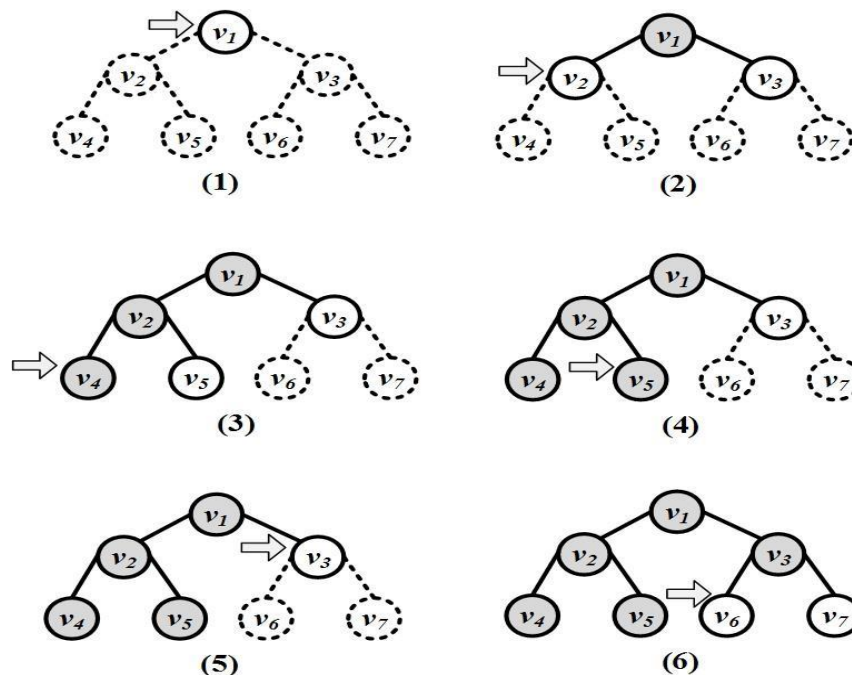
- We are interested in using a crawler that would collect relevant articles on a topic
 - Start from a page, e.g., Wikipedia
 - Employ some traversal technique to reach other relevant sites
- The main techniques:
 - **Depth-First Search (DFS)**
 - **Breadth-First Search (BFS)**
 - **Random Walk**



Source: stackoverflow

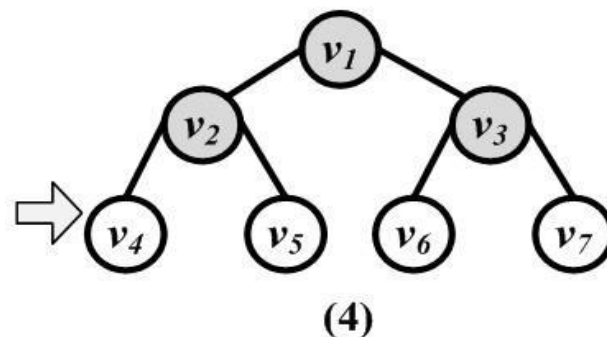
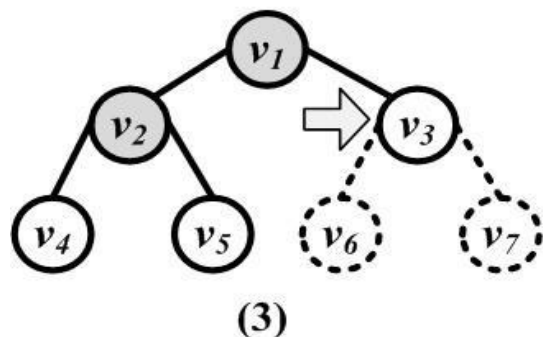
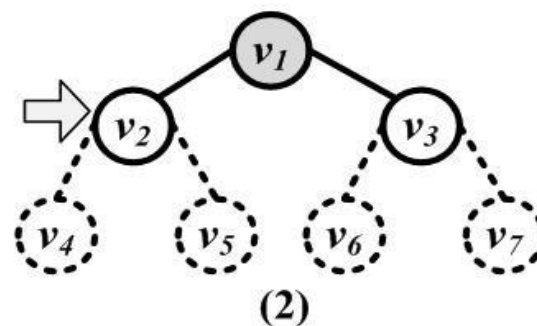
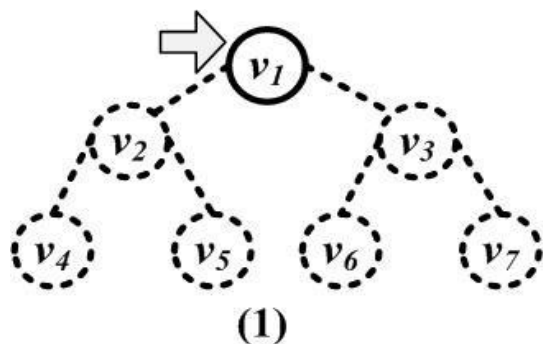
Depth-First Search (DFS)

- Depth-First Search (DFS) starts from a node v_i , selects one of its neighbors v_j from $N(v_i)$ and performs Depth-First Search on v_j before visiting other neighbors in $N(v_i)$



Breadth-First Search (BFS)

- BFS starts from a node and visits all its immediate neighbors first, and then moves to the second level by traversing their neighbors.



Finding Minimum Spanning Tree

Prim's Algorithm: Finding Minimum Spanning Tree

Finds MST in a weighted graph

1. Selecting a random node and add it to the MST
2. Grows the spanning tree by selecting edges which have one endpoint in the existing spanning tree and one endpoint among the nodes that are not selected yet. Among the possible edges, the one with the minimum weight is added to the set (along with its end-point).
3. This process is iterated until the graph is fully spanned

Prim's Algorithm Execution Example

